

LAB EXERCISE 2

AIM:

To add the constraints like primary key, foreign key, unique key and check using DDL commands

Description:

PRIMARY KEY:

The PRIMARY KEY constraint uniquely identifies each record in a database table.

Primary keys must contain UNIQUE values, and cannot contain NULL values.

A table can have only one primary key, which may consist of single or multiple fields.

FOREIGN KEY:

➤ A FOREIGN KEY is a key used to link two tables together.

➤ A FOREIGN KEY is a field (or collection of fields) in one table that refers to the PRIMARY KEY in another table.

➤ The table containing the foreign key is called the child table, and the table containing the candidate key is called the referenced or parent table.

UNIQUE Constraint:

➤ The UNIQUE constraint ensures that all values in a column are different.

➤ Both the UNIQUE and PRIMARY KEY constraints provide a guarantee for uniqueness for a column or set of columns.

➤ A PRIMARY KEY constraint automatically has a UNIQUE constraint.

➤ However, you can have many UNIQUE constraints per table, but only one PRIMARY KEY

constraint per table.

CHECK Constraint:

➤ The CHECK constraint is used to limit the value range that can be placed in a column

➤ If you define a CHECK constraint on a single column it allows only certain values for this column.

➤ If you define a CHECK constraint on a table it can limit the values in certain columns based on

1) SUPPLIER TABLE

```
1  CREATE TABLE Supplier(  
2  | Supplier_ID INT,  
3  | Supplier_Name VARCHAR(30)  
4  );  
5  INSERT INTO Supplier VALUES  
6  (1, 'Fresh Farms'),  
7  (2, 'Green Valley'),  
8  (3, 'Organic Hub'),  
9  (4, 'Nature Fresh'),  
10 (5, 'Healthy Harvest');  
11  
12 SELECT * FROM Supplier;
```

Output:

Supplier_ID	Supplier_Name
1	Fresh Farms
2	Green Valley
3	Organic Hub
4	Nature Fresh
5	Healthy Harvest

2) CATEGORY TABLE

```
14 CREATE TABLE Category(  
15 | Category_ID INT,  
16 | Category_Name VARCHAR(30)  
17 );  
18 INSERT INTO Category VALUES  
19 (101, 'Leafy Vegetables'),  
20 (102, 'Root Vegetables'),  
21 (103, 'Fruit Vegetables'),  
22 (104, 'Bulb Vegetables'),  
23 (105, 'Cruciferous');  
24  
25 SELECT * FROM Category;
```

Category_ID	Category_Name
101	Leafy Vegetables
102	Root Vegetables
103	Fruit Vegetables
104	Bulb Vegetables
105	Cruciferous

3)VEGETABLE SUPERMARKET DETAILS:

```
CREATE TABLE Vegetable_Supermarket(
    Veg_ID INT,
    Veg_Name VARCHAR(20),
    Category VARCHAR(30),
    Price DECIMAL(6,2),
    Quantity INT,
    Supplier_ID INT
);
INSERT INTO Vegetable_Supermarket VALUES
(1001,'Tomato','Fruit Vegetables',40.50,100,1),
(1002,'Potato','Root Vegetables',30.00,200,2),
(1003,'Carrot','Root Vegetables',55.00,120,3),
(1004,'Onion','Bulb Vegetables',45.00,180,4),
(1005,'Cabbage','Cruciferous',35.00,90,5);

SELECT * FROM Vegetable_Supermarket;
```

Veg_ID	Veg_Name	Category	Price	Quantity	Supplier_ID
1001	Tomato	Fruit Vegetables	40.50	100	1
1002	Potato	Root Vegetables	30.00	200	2
1003	Carrot	Root Vegetables	55.00	120	3
1004	Onion	Bulb Vegetables	45.00	180	4
1005	Cabbage	Cruciferous	35.00	90	5

Q1)

```
ALTER TABLE Vegetable_Supermarket
ADD PRIMARY KEY(Veg_ID);
```

Field	Type	Null	Key	Default	Extra
Veg_ID	int	NO	PRI	NULL	
Veg_Name	varchar(20)	YES		NULL	
Category	varchar(30)	YES		NULL	
Price	decimal(6,2)	YES		NULL	
Quantity	int	YES		NULL	
Supplier_ID	int	YES		NULL	

Q2)

```
ALTER TABLE Supplier
ADD PRIMARY KEY(Supplier_ID);
```

```
ALTER TABLE Supplier
ADD CONSTRAINT UQ_Supplier UNIQUE(Supplier_Name);
```

Field	Type	Null	Key	Default	Extra
Supplier_ID	int	NO	PRI	NULL	
Supplier_Name	varchar(30)	YES	UNI	NULL	

Q3)

```
ALTER TABLE Category
ADD PRIMARY KEY(Category_ID);
```

Field	Type	Null	Key	Default	Extra
Category_ID	int	NO	PRI	NULL	
Category_Name	varchar(30)	YES		NULL	

Q4)

```
ALTER TABLE Vegetable_Supermarket
ADD CONSTRAINT CHK_Price
CHECK(Price>0);
```

```
ALTER TABLE Vegetable_Supermarket
ADD CONSTRAINT CHK_Quantity
CHECK(Quantity>0);
```

Q5)

```
ALTER TABLE Vegetable_Supermarket
ADD CONSTRAINT UQ_VegName
UNIQUE(Veg_Name);
```

Field	Type	Null	Key	Default	Extra
Veg_ID	int	NO	PRI	NULL	
Veg_Name	varchar(20)	YES	UNI	NULL	
Category	varchar(30)	YES		NULL	
Price	decimal(6,2)	YES		NULL	
Quantity	int	YES		NULL	
Supplier_ID	int	YES		NULL	

Q6)

```
ALTER TABLE Vegetable_Supermarket
ADD CONSTRAINT FK_Supplier
FOREIGN KEY(Supplier_ID)
REFERENCES Supplier(Supplier_ID);
```